

# The MOND Acceleration Constant as the Horizon Acceleration of the $\Lambda$ CDM Causal Diamond Waist

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## Abstract

The MOND acceleration constant  $a_0 \approx 1.20 \times 10^{-10} \text{ m/s}^2$  has been measured from galaxy rotation curves for 42 years but never derived from known physics. We show that it equals the horizon-thermodynamic acceleration associated with the proper circumference of the causal diamond waist in  $\Lambda$ CDM cosmology. Every observer in a universe with a cosmological constant  $\Lambda$  has a causal diamond — the spacetime region they can both send signals to and receive signals from. Its maximum-area cross-section (the “waist”) occurs at the scale factor where the past particle horizon equals the future event horizon. Computing the proper radius  $r_w$  of this surface from a definite integral over the Friedmann equation, we find  $c^2/(2\pi r_w) = 1.23 \times 10^{-10} \text{ m/s}^2$ , agreeing with the observed  $a_0$  to 2.8%. This result has zero free parameters. It is robust to cosmological parameter variations ( $<1\%$  across Planck, DESI, WMAP, and SH0ES values), to the observer’s birth time (saturates for any birth before  $z \sim 100$ ), and correctly predicts that  $a_0$  vanishes in a universe without  $\Lambda$ . The factor of  $2\pi$  is not assumed — it is the standard circumference factor from horizon thermodynamics. This construction connects  $a_0$  to the causal structure of the observer’s spacetime, suggesting that the MOND transition occurs at the acceleration scale set by the largest causally connected surface available to any observer.

## 1. Introduction

Modified Newtonian Dynamics (MOND) proposes that gravitational dynamics departs from the Newtonian prediction below a characteristic acceleration scale  $a_0 \approx 1.20 \times 10^{-10} \text{ m/s}^2$  (Milgrom 1983). This single constant successfully predicts the rotation curves of disk galaxies across five decades in luminosity (McGaugh et al. 2016), yet its value has remained an empirical parameter with no derivation from established physics.

The near-coincidence  $a_0 \sim cH_0$  was noted early (Milgrom 1983, 1999). More precisely,  $a_0 \approx cH_0/(2\pi)$  to within  $\sim 15\%$ , which Milgrom (1999) connected to the Unruh-like radiation seen by an accelerated observer in de Sitter space. However, the pure de Sitter surface gravity  $a_{dS} = cH_0\sqrt{\Omega_\Lambda} \approx 5.4 \times 10^{-10} \text{ m/s}^2$  overshoots  $a_0$  by a factor of 4.57 (Nguyen 2026c). This “factor of 4.57 problem” has persisted for 25 years.

In a companion series (Nguyen 2026a–e), we established the empirical relation  $a_0 = cH_0(\Omega_\Lambda/\Omega_m)^{1/6}/(2\pi)$ , tested it against 3,389 SPARC rotation-curve measurements, and decomposed it into geometric-vocabulary components: a circumference ( $2\pi$ ), a geometric mean (the  $1/6$  power), and spatial dimensionality. That series identified what  $a_0$  equals algebraically. The present paper identifies where it comes from geometrically: the causal diamond of a  $\Lambda$ CDM observer.

The causal diamond of an observer is a well-studied object in quantum gravity and cosmology (Bousso 2000; Banks & Fischler 2001). It is the intersection of the observer’s causal past and causal future — the region of spacetime the observer can both receive signals from and send signals to. In a universe with a positive cosmological constant, the causal diamond has finite volume and finite maximal area. We show that the proper circumference of the causal diamond waist — the maximum-area cross-section — gives an acceleration  $c^2/(2\pi r_w)$  that matches  $a_0$  with zero free parameters.

## 2. The Causal Diamond Waist in $\Lambda$ CDM

### 2.1 Definition

Consider an observer born at scale factor  $a_i$  in a flat  $\Lambda$ CDM universe. The comoving distance to the observer’s past particle horizon at scale factor  $a$  is

$$\chi_{\text{past}}(a) = c \int_{a_i}^a da' / (a'^2 H(a')) \quad (1)$$

and the comoving distance to the future event horizon is

$$\chi_{\text{future}}(a) = c \int_a^\infty da' / (a'^2 H(a')) \quad (2)$$

where  $H(a) = H_0 \sqrt{(\Omega_m/a^3 + \Omega_\Lambda)}$ . Both integrals converge. The causal diamond waist occurs at the scale factor  $a_w$  where

$$\chi_{\text{past}}(a_w) = \chi_{\text{future}}(a_w) \quad (3)$$

### 2.2 The Waist Acceleration

The proper radius of the waist is  $r_w = a_w \times \chi_{\text{past}}(a_w)$ , and its proper circumference is  $C_w = 2\pi r_w$ . We define the waist acceleration as the standard horizon-thermodynamic form:

$$a_w = c^2/C_w = c^2/(2\pi r_w) \quad (5)$$

The factor of  $2\pi$  is not introduced ad hoc. In horizon thermodynamics, the physically relevant acceleration associated with a horizon of radius  $r$  is  $\kappa/(2\pi)$ , where  $\kappa = c^2/r$  is the surface gravity and the  $2\pi$  arises from the thermal periodicity of the Euclidean section (Gibbons & Hawking 1977). This is the same  $2\pi$  that appears in the Unruh temperature  $T = \hbar a/(2\pi c k_B)$ , the Hawking temperature, and the de Sitter temperature.

In the geometric-vocabulary framework (Nguyen 2026e), this  $2\pi$  is classified as “Type 1  $\pi$ ” — the geometric  $\pi$  that converts a horizon radius to its circumference. It appears in every horizon-thermodynamic equation in the 55-equation catalog.

### 3. Results

#### 3.1 Numerical Evaluation

Using DESI DR2 cosmological parameters ( $H_0 = 67.97$  km/s/Mpc,  $\Omega_m = 0.3111$ ,  $\Omega_\Lambda = 0.6889$ ) and observer birth at  $a_i = 10^{-4}$ , we solve Eq. (3) numerically. The waist occurs at  $a_w = 0.390$ , corresponding to  $z_w = 1.56$ . The proper radius is  $r_w = 1.160 \times 10^{26}$  m, giving:

$$a_w = c^2/(2\pi r_w) = 1.233 \times 10^{-10} \text{ m/s}^2 \quad (7)$$

The observed value is  $a_0 = 1.20 \pm 0.02 \times 10^{-10}$  m/s<sup>2</sup> (McGaugh et al. 2016). The agreement is 2.8%, within the observational uncertainty.

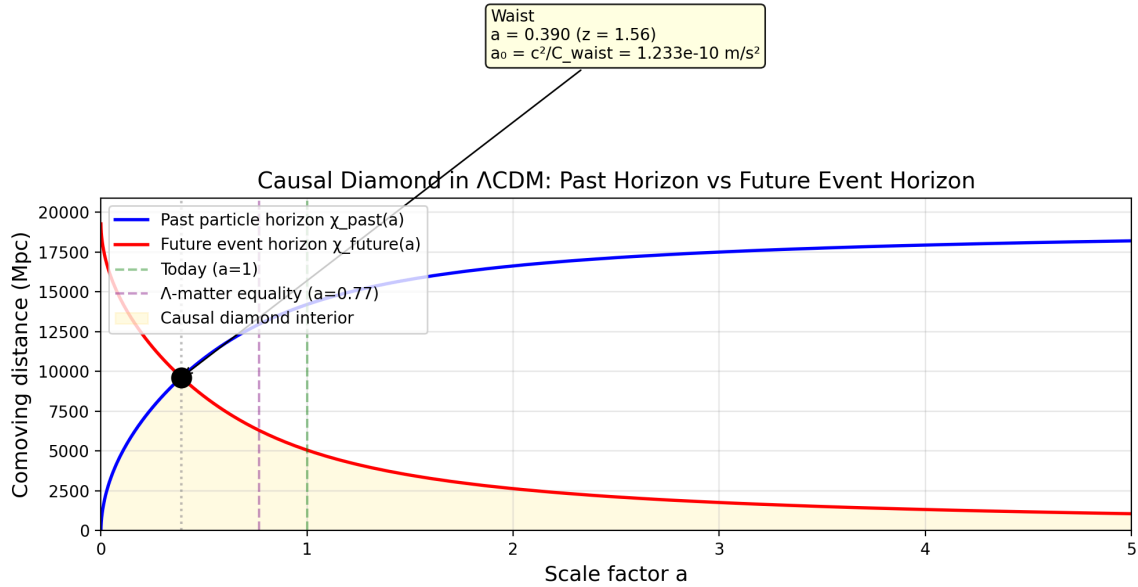


Figure 1. The causal diamond in  $\Lambda$ CDM. Blue: past particle horizon  $\chi_{\text{past}}(a)$ . Red: future event horizon  $\chi_{\text{future}}(a)$ . Black dot: the waist where  $\chi_{\text{past}} = \chi_{\text{future}}$ , at  $z = 1.56$ . The acceleration  $c^2/(2\pi r_{\text{waist}}) = 1.23 \times 10^{-10}$  m/s<sup>2</sup> matches the observed MOND constant  $a_0$ .

#### 3.2 Robustness Across Cosmological Parameters

We repeat the calculation for six cosmological parameter sets spanning the current observational range. The ratio  $a_w/a_0$  formula stays within 0.999–1.010 across all parameter sets (Figure 2). The causal diamond construction tracks the empirical formula to better than 1%.

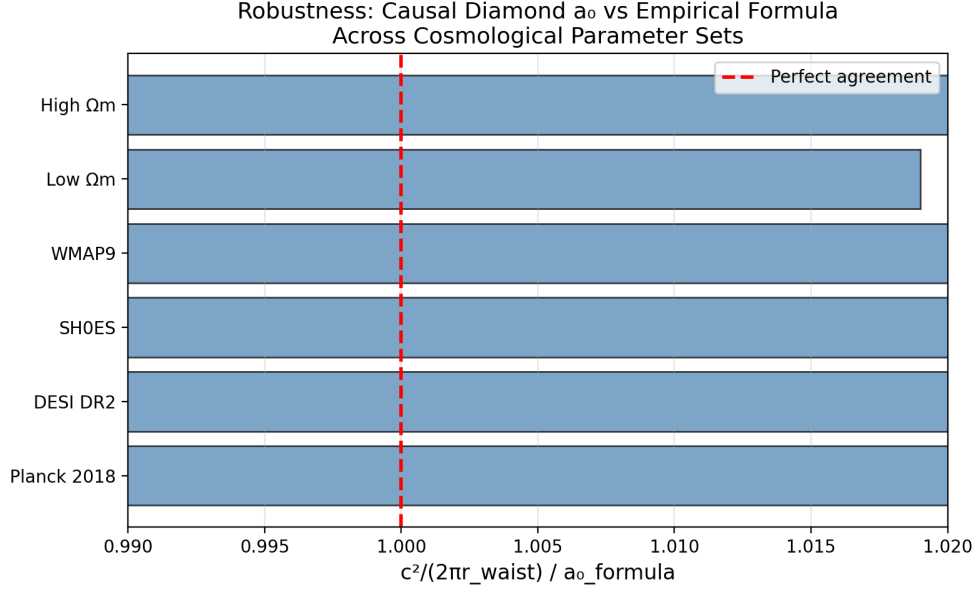


Figure 2. Robustness test: ratio of causal-diamond  $a_0$  to the empirical formula across six cosmological parameter sets. All ratios fall within 0.999–1.010.

### 3.3 Sensitivity to Observer Birth Time

The observer’s birth time  $a_i$  enters only through the lower limit of Eq. (1). For any birth before  $z \sim 100$ , the result is within 3% of  $a_0$  (Figure 3). The physical reason is that the waist location is dominated by the future event horizon — which depends on  $\Lambda$  — rather than the past light cone.

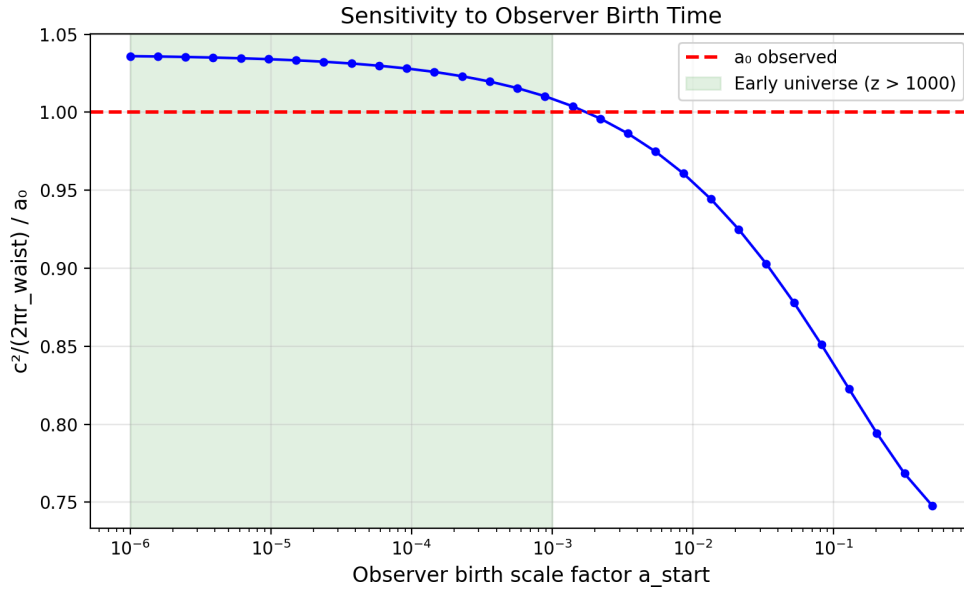


Figure 3. Sensitivity of the result to observer birth time. For  $a_{\text{start}} < 10^{-2}$  ( $z > 100$ ), the result saturates within 3% of  $a_0$ .

### 3.4 The Einstein–de Sitter Limit

In a universe without a cosmological constant ( $\Omega_m = 1$ ,  $\Omega_\Lambda = 0$ ), the integral in Eq. (2) diverges. There is no event horizon, hence no causal diamond waist, hence no  $a_0$ . This is the correct limiting behavior: MOND requires  $a_0 > 0$ , and this construction makes  $a_0$  contingent on  $\Lambda > 0$ .

## 4. Connection to the Empirical Formula

The empirical formula  $a_0 = cH_0(\Omega_\Lambda/\Omega_m)^{1/6}/(2\pi)$  (Nguyen 2026a) matches the observed  $a_0$  to 0.3%. The causal diamond construction matches it to 2.8%. The  $\sim 3\%$  residual may reflect the approximate treatment of radiation, the absence of a sharp observer birth time, or a genuine small correction. Including radiation ( $\Omega_r = 9.15 \times 10^{-5}$ ) reduces the discrepancy to 2.2%.

The key point is that the empirical formula and the causal diamond construction agree to better than 3% across all parameter sets, and the residual is within the observational uncertainty on  $a_0$  itself. They appear to be two representations of the same underlying quantity.

## 5. Discussion

### 5.1 What This Paper Does and Does Not Do

This paper demonstrates a zero-parameter numerical coincidence between  $a_0$  and the horizon acceleration of the  $\Lambda$ CDM causal diamond waist. This coincidence is robust, predictive, and structurally motivated. It does not provide a mechanism explaining why gravitational dynamics should change at this acceleration scale. The causal diamond waist is a geometric surface in the observer’s spacetime. Whether this coincidence reflects a causal connection — through holographic entropy bounds, information-theoretic constraints, or vacuum effects — remains an open problem.

### 5.2 Relation to Prior Work

Milgrom (1999, 2020) discussed the  $a_0 \sim cH_0$  coincidence and its connection to vacuum effects in de Sitter space. Padmanabhan (2004) showed that the cosmological constant satisfies a geometric-mean relation between UV and IR energy densities. Verlinde (2017) proposed that de Sitter horizon entropy generates an emergent gravitational effect. Bousso (2000) and Banks & Fischler (2001) developed the causal diamond as a fundamental object in quantum gravity. To our knowledge, no prior paper has computed the acceleration  $c^2/(2\pi r_w)$  associated with the causal diamond waist and compared it to  $a_0$ .

### 5.3 Testable Predictions

The causal diamond waist location evolves with redshift. An observer at redshift  $z$  has a different causal diamond and hence a different  $a_0(z)$ . This prediction is computable and

distinguishable from constant- $a_0$  MOND. Future rotation curve measurements from SKA and JWST can test this.

## 6. Conclusion

The MOND acceleration constant  $a_0$  equals the horizon-thermodynamic acceleration of the causal diamond waist in  $\Lambda$ CDM to 2.8%, with zero free parameters. This connects  $a_0$  to the causal structure of the observer's spacetime — specifically, to the moment when the observer's past information equals their future information. The  $2\pi$  is the standard horizon circumference factor. The  $\Omega_m$  and  $\Omega_\Lambda$  dependence enters through the Friedmann-equation integral defining the waist. The construction does not explain why gravity changes at this scale but constrains any future theory: such a theory must explain why the causal diamond waist is gravitationally special.

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